

Eden Danielsen–Jensen

Ph.D. Candidate in Mathematics

Department of Mathematics, Brigham Young University • Provo, Utah, USA

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Employment

Teaching Assistant Brigham Young University	2024–Present
Sessional Academic Tutor University of the Sunshine Coast	2023–2024

Education

Brigham Young University Ph.D. in Mathematics	Expected Graduation: August 2027
• Advisor: Jennifer Brooks. • Dissertation: Group-invariant CR Maps from Spheres to Hyperquadrics (working title).	
University of the Sunshine Coast Bachelor of Science (Honours), Mathematics — First Class Honours	2022
• Advisors: Robert McDougall & Lauren Thornton. • Honours thesis: <i>A Revised Introduction for the Base Semisimple Class</i>.	
University of the Sunshine Coast Bachelor of Science — GPA 6.96/7.00	2021
• Major: Mathematics • Minor: Chemistry	

Research Interests

Several Complex Variables; CR Geometry; proper holomorphic mappings; group-invariant CR maps; hyperquadrics; invariant polynomial maps; computational methods in complex analysis.

Publications

In Preparation

- Jennifer Brooks, Eden Danielsen–Jensen, Dusty Grundmeier, and Alex Lee. *Lower Bounds for Invariant CR Maps*. In preparation.
- Eden Danielsen–Jensen, Dusty Grundmeier, Abdullah Al Helal, Valentin Kunz, Jiří Lebl, Ming Xiao, and Weixia Zhu. *Degree-Three Rational Sphere Maps: Sharp Denominator Region and Gram Normal Forms*. In preparation; working title.

Presentations

Signatures for Group-Invariant Monomial Maps from Spheres to Hyperquadrics

• Joint Mathematics Meetings (JMM)	2026
• MAA Intermountain Section Meeting	2026
• BYU Student Research Conference	2026

A Revised Introduction for the Base Semisimple Class

• Australian Mathematical Society Conference	2022
• Australian Algebra Conference	2022

Research Activities

Collaborative Research Workshop in Several Complex Variables

2026

Institute for Advanced Study (IAS), Princeton NJ

Teaching

Instructor of Record — Brigham Young University

- **MATH 213** Elementary Linear Algebra Summer 2026
- **MATH 109** Precalculus Winter 2026
- **MATH 116** Essentials of Calculus Fall 2025
- **MATH 112** Calculus I Spring 2025

Teaching Assistant / Recitation Leader — Brigham Young University

- **MATH 341** Theory of Analysis I Fall 2025, Fall 2026
- **MATH 113** Calculus II Fall 2026

Grader — Brigham Young University

- **MATH 570** Matrix Analysis Winter 2025
- **MATH 510** Numerical Linear Algebra Fall 2025
- **MATH 314** Calculus of Several Variables Summer 2024

Sessional Academic Tutor — University of the Sunshine Coast

- **MATH 104** Introductory Calculus Semester 2 2023
- **MATH 103** Introduction to Applied Mathematics Semester 1 2023
- **SCI 110** Science Research Methods Semester 1 & 2 2023

Teaching Portfolio

A sample of course materials and resources that I have created is available on my website: eden-dj.github.io/ACADEMIC-WEBSITE/.

Service & Outreach

- Graduate Student Cohort Representative, BYU Mathematics 2024–Present
- Organizer, Graduate Student Writing Group, BYU Mathematics 2026–Present
- Organizer, Graduate Professional Development Workshops, BYU Mathematics 2025–Present
- Graduate School Q&A Panelist, BYU Mathematics 2025, 2026
- Conference Organization Support, Australian Algebra Conference 2022

Honors & Awards

- Vice-Chancellor's Honours Scholarship, University of the Sunshine Coast 2022
- Thompson Excellence Scholarship, University of the Sunshine Coast 2019–2021
- Rural and Regional Enterprise Scholarship 2019–2021

Professional Memberships

- American Mathematical Society Member
- Mathematical Association of America Member
- Australian Mathematical Society Member
- Women in Mathematics Special Interest Group (WIMSIG) AustMS Member